Code No: R07A1EC10 **R07** **Set No. 2**

I B.Tech Examinations,December/January -2011/2012
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Write an example to explain about call by value.
   (b) Write a factorial program using no return and no argument type. [10+6]
2. Explain need of pointers and its advantage. [16]
3. (a) Explain in detail about different data types available in C.
   (b) Write the structure of a 'C' program and explain. [8+8]
4. (a) Find the value of sec 34^0 given the following data

θ:	31^0	32^0	33^0	34^0
tan θ:	0.6008	0.6249	0.6494	0.6745

   (b) Interpolate y at x = 5 from the following data [8+8]

X	1	2	3	4	7
Y	2	4	8	16	128

5. (a) Evaluate $\int_1^6 \frac{dx}{1+x^4}$ by Trapezoidal rule and Simpson's 1/3rd rule.
   (b) Using Euler's method find y (0.2) given dy/dx = log(x + y) and y (0) = 1, h = 0.2. [8+8]
6. What is a queue? Explain the various operations in queue using a C program. [16]
7. Create a structure for bank using following members name, address, balance and check whether the balance is within the range.
   (a) 5000 to 10,000
   (b) 10,001 to 20,000
   (c) > 20,000. [16]
8. (a) Find a positive root of the following equation by bisection method 3x-1=cosx.
   (b) Solve by iteration method: 3x+sinx=e^x. [8+8]

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Code No: R07A1EC10 **R07** **Set No. 4**

I B.Tech Examinations,December/January -2011/2012
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. What is a queue? Explain the various operations in queue using a C program. [16]

2. (a) Find a positive root of the following equation by bisection method 3x-1=cosx.
   (b) Solve by iteration method: 3x+sinx=e^x. [8+8]

3. (a) Explain in detail about different data types available in C.
   (b) Write the structure of a 'C' program and explain. [8+8]

4. (a) Evaluate $\int_1^6 \frac{dx}{1+x^4}$ by Trapezoidal rule and Simpson's 1/3rd rule.
   (b) Using Euler's method find y (0.2) given dy/dx = log(x + y) and y (0) = 1, h = 0.2. [8+8]

5. Explain need of pointers and its advantage. [16]

6. (a) Write an example to explain about call by value.
   (b) Write a factorial program using no return and no argument type. [10+6]

7. Create a structure for bank using following members name, address, balance and check whether the balance is within the range.
   (a) 5000 to 10,000
   (b) 10,001 to 20,000
   (c) > 20,000. [16]

8. (a) Find the value of sec 34^0 given the following data

θ:	31^0	32^0	33^0	34^0
tan θ:	0.6008	0.6249	0.6494	0.6745

   (b) Interpolate y at x = 5 from the following data [8+8]

X	1	2	3	4	7
Y	2	4	8	16	128

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Code No: R07A1EC10 **R07** **Set No. 1**

I B.Tech Examinations,December/January -2011/2012
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. What is a queue? Explain the various operations in queue using a C program. [16]

2. (a) Evaluate $\int_1^6 \frac{dx}{1+x^4}$ by Trapezoidal rule and Simpson's 1/3rd rule.
   (b) Using Euler's method find y (0.2) given dy/dx = log(x + y) and y (0) = 1, h = 0.2. [8+8]

3. (a) Find the value of sec 34^0 given the following data

θ:	31^0	32^0	33^0	34^0
tan θ:	0.6008	0.6249	0.6494	0.6745

   (b) Interpolate y at x = 5 from the following data [8+8]

X	1	2	3	4	7
Y	2	4	8	16	128

4. (a) Find a positive root of the following equation by bisection method 3x-1=cosx.
   (b) Solve by iteration method: 3x+sinx=e^x. [8+8]

5. Explain need of pointers and its advantage. [16]

6. (a) Write an example to explain about call by value.
   (b) Write a factorial program using no return and no argument type. [10+6]

7. (a) Explain in detail about different data types available in C.
   (b) Write the structure of a 'C' program and explain. [8+8]

8. Create a structure for bank using following members name, address, balance and check whether the balance is within the range.
   (a) 5000 to 10,000
   (b) 10,001 to 20,000
   (c) > 20,000. [16]

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Code No: R07A1EC10 **R07** **Set No. 3**

I B.Tech Examinations,December/January -2011/2012
COMPUTER PROGRAMMING AND NUMERICAL METHODS
Metallurgy And Material Technology

Time: 3 hours **Max Marks: 80**

Answer any FIVE Questions
All Questions carry equal marks

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1. (a) Explain in detail about different data types available in C.
   (b) Write the structure of a 'C' program and explain. [8+8]
2. What is a queue? Explain the various operations in queue using a C program. [16]
3. (a) Find a positive root of the following equation by bisection method 3x-1=cosx.
   (b) Solve by iteration method: 3x+sinx=e^x. [8+8]
4. (a) Write an example to explain about call by value.
   (b) Write a factorial program using no return and no argument type. [10+6]
5. Create a structure for bank using following members name, address, balance and check whether the balance is within the range.
   (a) 5000 to 10,000
   (b) 10,001 to 20,000
   (c) > 20,000. [16]
6. (a) Find the value of sec 34^0 given the following data

θ:	31^0	32^0	33^0	34^0
tan θ:	0.6008	0.6249	0.6494	0.6745

   (b) Interpolate y at x = 5 from the following data [8+8]

X	1	2	3	4	7
Y	2	4	8	16	128

7. Explain need of pointers and its advantage. [16]
8. (a) Evaluate $\int_1^6 \frac{dx}{1+x^4}$ by Trapezoidal rule and Simpson's 1/3rd rule.
   (b) Using Euler's method find y (0.2) given dy/dx = log(x + y) and y (0) = 1, h = 0.2. [8+8]

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